HOSTEL MANAGEMENT SYSTEM

SOFTWARE ARCHITECTURE & DESIGN SPECIFICATION

PROJECT TITLE	Hostel Management System (HMS)
DOCUMENT TITLE	Software Architecture & Design Specification
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1. Introduction

1.1 Purpose

This document specifies the proposed software architecture and system-level design for the Hostel Management System (HMS). It relates the functional requirements, non-functional requirements, security objectives, security requirements, and business rules defined in the Software Requirements Specification (SRS) to an architectural model. The system is intended to support routine hostel administration activities: room allocation, fee viewing and payment, complaint handling, visitor logging, weekly mess menus, leave/outpass processing, notices, and facility booking.

The document describes the proposed structural decomposition, logical layering, component dependencies, requirements traceability, security architecture, and a conceptual deployment view. It does not commit to specific programming languages, frameworks, servers, or database products.

1.2 Scope

The SRS describes HMS as a browser-based application accessed through regular web browsers, with a relational database server at the backend. The system is intended to support three user roles:

- **Student:** view own profile, view and pay hostel fees, register and track complaints, submit leave/outpass requests, view the weekly mess menu, read notices, and view and book facilities.
- **Warden:** log visitor entry and exit, and approve or reject leave/outpass requests.
- **Administrator:** manage student profiles, allocate rooms, publish the weekly mess menu, and post notices.

This document covers the proposed architecture pattern and rationale, component decomposition and dependencies, traceability to the SRS, security architecture, quality attribute considerations, the placement of business rule enforcement, and a conceptual deployment view.

1.3 Document Conventions

- **SRS identifiers:** FR-xx (Functional Requirements), NFR-xx (Non-Functional Requirements), SO-xx (Security Objectives), SR-xx (Security Requirements), BR-xx (Business Rules), and UC-xx (Use Cases).
- **Architectural stance:** This is an architecture specification. It describes a proposed design; it makes no claim that the system has been built, tested, or deployed.
- **Technology independence:** Only structural roles (for example, Web Interface, Relational Database) are named. No technology stack is specified.

1.4 References

- ISO/IEC/IEEE 29148:2018, Systems and software engineering - Life cycle processes - Requirements engineering.
- Hostel Management System (HMS) Software Requirements Specification (SRS), prepared by Member 1.

2. Architecture Overview

2.1 Architectural Goals

The goal is a modular and maintainable structure that is designed to support the functional requirements, quality attributes, security objectives, and business rules in the SRS. The proposed structure separates user interaction, business logic, and data storage, and supports coordination between the Student, Warden, and Administrator roles.

2.2 Proposed Layered Architecture

The SRS does not specify an architecture pattern. The proposed architecture adopts a layered structure to organize the system into presentation, application/business, data access, and database layers. This is a proposed design

decision of this document, not a requirement of the SRS. The four layers are:

- **1. Presentation / Web Interface Layer:** provides the browser-based screens through which Student, Warden, and Administrator users interact with the system, and passes user requests to the Application / Business Layer.
- **2. Application / Business Layer:** contains the functional components (Authentication, Student, Room, Fee, Complaint, Visitor, Mess, Leave / Outpass, Notice, and Facility Booking). It is responsible for role-based access checks and for enforcing the business rules associated with each module.
- **3. Data Access Layer:** mediates between the Application / Business Layer and the Relational Database, so that business components do not depend directly on how data is stored.
- **4. Relational Database:** provides persistent storage for the data required by HMS.

2.3 Architectural Rationale

- **Separation of concerns:** keeping user interaction, business rules, and data access apart limits the impact of a change in one layer on the others.
- **Maintainability:** each functional component has a distinct responsibility, so each business function (for example Fee Management or Room Management) can be understood and reviewed on its own against its business rules.
- **Centralized security and rules:** role checks and business rule enforcement are placed in the Application / Business Layer, so they apply regardless of which screen initiates a request.
- **Technology independence:** the Data Access Layer boundary keeps business components independent of storage details.

3. Component Architecture

3.1 UML Component Diagram

HMS is accessed by three SRS-defined roles: Student, Warden, and Administrator, through the Presentation / Web Interface. The diagram shows the proposed components and their logical dependencies across the layered architecture. Dependencies flow downward: Presentation uses the Application / Business components, which use Data Access, which uses the Relational Database.

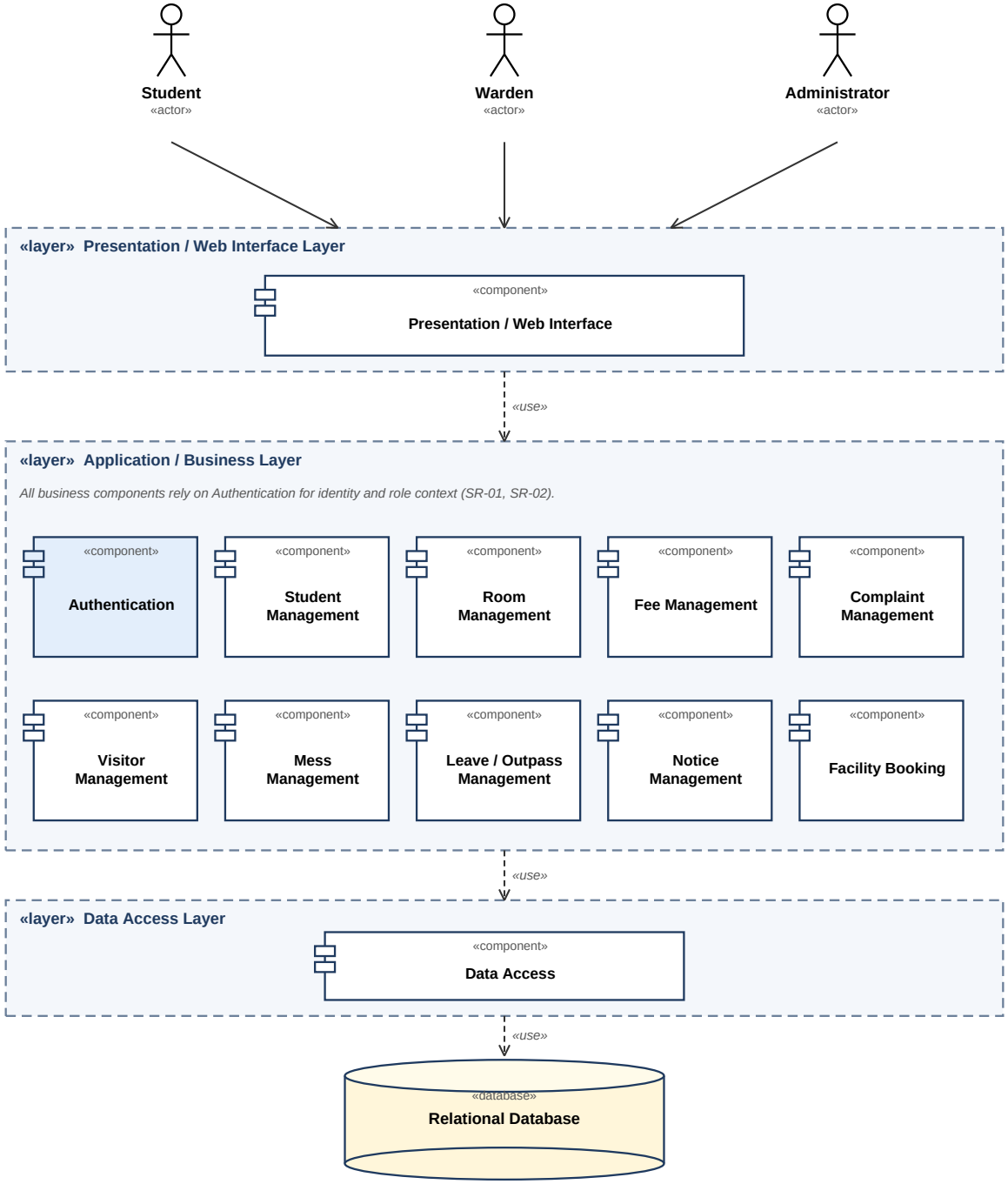

Figure 3.1: UML-style component diagram of the Proposed Layered Architecture

3.2 Component Descriptions

Table 3.1 lists the responsibilities and related SRS requirements of each proposed component.

Table 3.1: Component Descriptions and Requirement Traceability

Component	Responsibility	Related Requirements
Presentation / Web Interface	Provides browser-based interaction for Student, Warden, and Administrator roles and passes requests to the business components.	NFR-04
Authentication	Authenticates Administrator, Warden, and Student users with hashed passwords and establishes the role context used for role-based access control.	UC-03, SR-01, SR-02
Student Management	Maintains student profile information. Administrator creates and updates records; a Student views their own profile.	FR-02
Room Management	Supports room allocation by authorized staff. Provides the place for enforcing that a student cannot have more than one active room.	FR-01, SO-02, BR-01
Fee Management	Maintains fee and payment records. Supports fee viewing and partial payments; a fee is Paid only when the outstanding balance is zero. Modification is restricted to authorized staff.	FR-03, SO-01, SO-02, BR-03
Complaint Management	Supports complaint registration and status tracking. Provides the place for enforcing the sequence Open -> In Progress -> Resolved.	FR-04, BR-04
Visitor Management	Supports Warden logging of visitor entry and exit. Exit timestamp may be null while the visitor remains on the premises.	FR-05, BR-05
Mess Management	Supports Administrator publication of the weekly mess menu and viewing by students.	FR-06
Leave / Outpass Management	Supports student submission of leave/outpass requests and their approval or rejection by the Warden.	FR-07, BR-02
Notice Management	Supports Administrator posting of notices for students to view.	FR-08
Facility Booking	Supports viewing facility availability and booking. Provides the place for rejecting a booking for a time slot that is already booked.	FR-09, BR-06
Data Access	Mediates persistence and retrieval between the business components and the Relational Database.	Supports all data-related requirements
Relational Database	Provides persistent storage for HMS data.	Supports all data-related requirements

4. Architecture-to-Requirements Traceability

Table 4.1 shows how the proposed architecture is intended to support each SRS requirement, objective, and business rule. It shows architectural support only; it does not claim that any requirement has been tested or achieved.

Table 4.1: Architecture-to-Requirements Traceability Matrix

Req ID	Requirement Summary	Component / Layer	How the Architecture Supports It
FR-01	Room allocation	Room Management	Provides the component through which authorized staff allocate rooms.
FR-02	Student profile management	Student Management	Provides profile maintenance by the Administrator and profile viewing by the Student.
FR-03	Fee viewing / payment	Fee Management	Provides fee viewing and payment recording, including partial payments.
FR-04	Complaint registration and tracking	Complaint Management	Provides complaint registration and status tracking.

Req ID	Requirement Summary	Component / Layer	How the Architecture Supports It
FR-05	Visitor entry/exit logging	Visitor Management	Provides Warden logging of visitor entry and exit.
FR-06	Weekly mess menu	Mess Management	Provides menu publication and student viewing.
FR-07	Leave/outpass submission and approval	Leave / Outpass Management	Provides student submission and Warden approval or rejection.
FR-08	Notices	Notice Management	Provides notice posting and viewing.
FR-09	Facility availability and booking	Facility Booking	Provides availability viewing and booking.
NFR-01	Response time	Presentation / Application layers	Separation of interface and business processing is designed to support the response-time requirement.
NFR-02	Concurrent users	Overall layered architecture	The layered structure is designed to support concurrent student use.
NFR-03	Availability	Overall architecture and database support	The layered structure and central database are designed to support the availability requirement, excluding scheduled maintenance.
NFR-04	Usability	Presentation / Web Interface	Role-oriented browser screens are designed for use by staff without technical training.
SO-01	Protect personal and payment data	Authentication, RBAC, Data Protection	Access to student personal and payment data is restricted by identity and role.
SO-02	Only authorized staff modify room and fee records	RBAC, Authorization	Room allocation and fee modification are restricted to authorized staff by role-based access control.
SR-01	Hashed-password authentication	Authentication, Password Protection	Authentication is designed around stored hashed passwords for all three roles.
SR-02	Role-based access control	RBAC, Authorization	Role context (Student, Warden, Administrator) determines permitted operations.
BR-01	One active room per student	Room Management	Application / Business Layer provides the place to enforce this rule.
BR-02	Leave/outpass needs Warden approval	Leave / Outpass Management	Approval is provided as a Warden operation.
BR-03	Partial payments; Paid at zero balance	Fee Management	Fee logic supports partial payment and marks Paid at zero balance.
BR-04	Open -> In Progress -> Resolved	Complaint Management	Complaint status handling follows the specified sequence.
BR-05	Entry and exit timestamps; exit may be null	Visitor Management	Visitor logging records entry and exit; exit may be null while on premises.
BR-06	No booking of a slot already booked	Facility Booking	Booking logic rejects a booking for an already booked time slot.

5. Security Architecture

The proposed security architecture is intended to support SO-01, SO-02, SR-01, and SR-02. Security checks are placed in the Application / Business Layer so that identity and role are checked before protected operations and data are accessed.

Table 5.0: Security architecture overview

Section	Focus	Supports
5.1 Authentication	Verify user identity before access	SR-01
5.2 Password Protection	Store and compare passwords as hashes	SR-01

Section	Focus	Supports
5.3 Role-Based Access Control	Distinguish Student, Warden, Administrator	SR-02
5.4 Authorization	Restrict room allocation and fee record modification to authorized staff	SO-02
5.5 Data Protection	Protect student personal and payment data	SO-01

5.1 Authentication

In line with SR-01, the Authentication component is designed to verify the identity of Administrator, Warden, and Student users before they can access protected functions. On successful authentication, the user's role is made available to the other components for role-based access control.

5.2 Password Protection

SR-01 requires authentication with hashed passwords. The architecture therefore treats stored passwords as hashed values rather than plaintext, and the Authentication component is designed to check submitted credentials against the stored hashed password.

5.3 Role-Based Access Control

In line with SR-02, role-based access control distinguishes the three roles:

- **Student:** self-service operations such as viewing own profile and fees, paying fees, registering complaints, submitting leave/outpass requests, viewing mess menus and notices, and booking facilities.
- **Warden:** visitor entry/exit logging and approval or rejection of leave/outpass requests.
- **Administrator:** student profile management, room allocation, mess menu publication, and notices.

The Application / Business Layer is designed to check the user's role before a component operation is performed.

5.4 Authorization

In line with SO-02, room allocation and fee record modification are restricted to authorized staff according to role-based access control. Students must not be allowed to modify these records; a student's fee-related access is limited to viewing and paying their own fees (FR-03).

5.5 Data Protection

In line with SO-01, the architecture is designed to protect student personal and payment data from unauthorized access or disclosure. Access to profile and payment data is controlled by authentication and role-based access control in the Application / Business Layer, and students are intended to see only their own records.

6. Quality Attribute Considerations

The proposed layered architecture is designed to support the following SRS quality requirements. No claim is made that any target has been measured or achieved.

6.1 Performance

SRS: NFR-01 (response time). **Consideration:** The layered structure is designed to support the response-time requirement NFR-01 by giving each layer a focused role.

6.2 Scalability

SRS: NFR-02 (concurrent users). **Consideration:** The layered structure is designed to support concurrent student use as required by NFR-02.

6.3 Availability

SRS: NFR-03 (availability, excluding scheduled maintenance). **Consideration:** The overall architecture and its central relational database are designed to support NFR-03.

6.4 Usability

SRS: NFR-04 (usable by staff without technical training). **Consideration:** The Presentation / Web Interface layer is intended to provide role-oriented, task-based screens to support NFR-04.

6.5 Security

SRS: SO-01, SO-02, SR-01, SR-02. **Consideration:** Authentication, password protection, role-based access control, authorization, and data protection (Section 5) are placed in the Application / Business Layer to support these requirements.

7. Business Rule Enforcement

The Application / Business Layer is responsible for enforcing the business rules associated with the corresponding functional modules. Because the rules sit in this layer, they are intended to apply consistently however a request originates.

Table 7.1: Business rule placement

Rule	SRS Business Rule	Responsible Component
BR-01	A student cannot have more than one active room.	Room Management (Application / Business Layer)
BR-02	Leave/outpass requires Warden approval.	Leave / Outpass Management
BR-03	Partial fee payments are allowed; a fee is Paid only when the outstanding balance is zero.	Fee Management
BR-04	Complaint status follows Open -> In Progress -> Resolved.	Complaint Management
BR-05	Visitor log requires entry and exit timestamps; exit may be null while the visitor remains on premises.	Visitor Management
BR-06	A facility cannot be booked if already booked for the same time slot.	Facility Booking

8. Deployment Architecture

The SRS establishes only that HMS is a browser-based application accessed through regular web browsers with a relational database server at the backend. The conceptual deployment view below reflects only this information.

Figure 8.1: Conceptual deployment view

- **User Device / Web Browser:** used by students, wardens, and administrators to access HMS.
- **Hostel Management System Application:** hosts the Web Interface, Application / Business Layer components, and Data Access Layer.
- **Relational Database Server:** provides persistent storage for HMS data.

9. Proposed Architectural Decisions

Table 9.1: Architectural decisions and rationale

ID	Proposed Decision	Rationale
AD-01	Adopt a layered architecture	The SRS does not prescribe an architecture pattern. A layered structure gives clear boundaries between presentation, business logic, data access, and storage.
AD-02	Centralize authentication and role-based access control	Placing them in the Application / Business Layer avoids repeating access checks in each module and supports SR-01 and SR-02.
AD-03	Place business rule enforcement in the Application / Business Layer	Keeps BR-01 to BR-06 with their functional modules rather than in the interface or storage.
AD-04	Separate data access from business components	A dedicated Data Access Layer keeps business components independent of storage details.
AD-05	Remain technology-neutral	The SRS specifies no programming language, framework, or database product, so none is chosen here.

10. Design Scope and Boundaries

This specification covers the Software Architecture responsibilities of Member 3. The SRS defines what the system must do (FR-01 to FR-09, NFR-01 to NFR-04, SO-01, SO-02, SR-01, SR-02, BR-01 to BR-06). This document proposes how the system may be logically organized to support them.

- **Included:** proposed layered architecture, component decomposition, UML-style component diagram, traceability, security architecture, quality attribute considerations, business rule placement, and conceptual deployment view.
- **Outside this document:** detailed class, sequence, and interface designs and source code, which belong to other team members and later phases.

11. Conclusion

The proposed architecture is designed to support the Hostel Management System's operational objectives while maintaining alignment with the requirements defined in the SRS. It organizes the system into Presentation / Web Interface, Application / Business, Data Access, and Relational Database layers.

The proposed components cover room allocation, student profiles, fees, complaints, visitor logging, mess menus, leave/outpass, notices, and facility booking. Authentication, password protection, role-based access control, authorization, and data protection provide the security architecture, and business rules BR-01 to BR-06 have a defined place in the Application / Business Layer. The document provides a basis for later design and implementation work, and makes no claim that the system has been implemented or verified.